

1946



238,000 MILES THE DISTANCE FROM EARTH TO THE MOON, AS SHOWN BY PROJECT DIANA

The distance between Earth and the Moon was measured by bouncing a radar signal off the Moon.

THE VERY FIRST CONTACT

WITH A SPACE BODY was made on January 10, when the US Army Signal Corps successfully detected the **echo of radar** reflected from the Moon—just 2.5 seconds after it was sent. **Project Diana** was initially established to examine whether long-range radar could detect incoming missiles, but it marked the start of the **US space program** as well. In addition to determining the distance to the

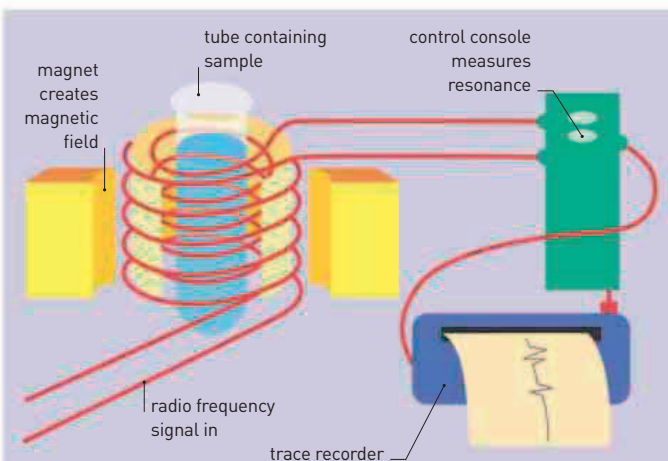
Moon, it demonstrated that signals made on Earth could be used to communicate in space.

The same month saw the publication of studies of a phenomenon that would revolutionize medical imaging: **nuclear magnetic resonance (NMR)**. Swiss physicists **Felix Bloch** (1905–83) and American physicist **Edward Purcell** (1912–97) showed that when a sample is exposed to an intense magnetic field, certain nuclei

within the sample resonate at characteristic frequencies. This technique was subsequently used to study the structures of molecules, and was later modified to produce images of bigger internal structures, such as those of the living body in **MRI (magnetic resonance imaging)**.

American biologists **Edward Tatum** (1909–75) and **Joshua Lederberg** (1925–2008) found that bacteria had a sexual process similar to that of more complex organisms. When they combined different strains, a few **bacteria** developed new abilities that the original strains were incapable of doing individually. They discovered that bacterial cells were binding together and exchanging genetic material in a process known as conjugation, thereby sharing chemical capabilities. This process has important implications, such as the spread of **antibiotic resistance**.

In July, the US undertook the **first post-war nuclear test** using the same sort of bomb as dropped on Nagasaki in 1945. Called **Test Able**, it was conducted on the Bikini Atoll in the Pacific Ocean, and was designed to see the effects on 78 experimental ships anchored in the lagoon—some carrying living test animals. More than 60 nuclear tests would later be carried out in the Bikini Atoll.



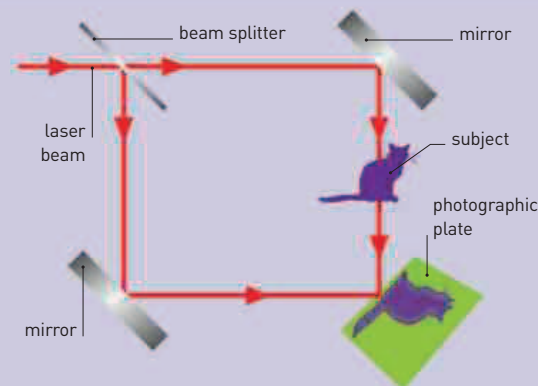
NUCLEAR MAGNETIC RESONANCE

The magnetic resonance of atomic nuclei can be used to detect the chemical components of a substance. The atomic nuclei in a test sample that has been placed in a strong magnetic field absorb and emit electromagnetic radiation at characteristic frequencies. Scientists can use this to establish the kinds of atoms present in the sample and determine its chemical structure.

1947



The Bell X-1 manned airplane was launched from the bomb bay of a Boeing B-29 to reach a record-breaking altitude and break the sound barrier.



HOLOGRAMS

Hologram technology involves directing a laser beam onto a photographic plate using mirrors. The beam is split so that part of it bounces off the subject before hitting the plate, while the other part hits the plate directly. This produces an interference pattern, which is photographed. A 3-D image is made by shining a laser onto the negative at the same angle as the original beam.

IN 1947, AMERICAN PHYSICIST LUIS ALVAREZ (see 1980)

oversaw the construction of the first **proton linear accelerator** in Berkley, California. In the same year, American **physicist William Hansen** (1909–49) produced the **first electron linear accelerator** at Stanford University, California. Particle acceleration is used to study the building blocks of matter, and it has practical applications, such as in the treatment of cancer.

In September at the American Chemical Society, Jacob

Marinsky (1918–2005) announced the **discovery of radioactive element number 61**, named **promethium** after the Greek god Prometheus. Promethium is a rare earth element, which closed the gap in the periodic table.

In October, US Air Force pilot Charles “Chuck” Yeager (b.1923) became the **first person to break the sound barrier**. He flew **Bell X-1** and achieved a speed of Mach 1.07 or 815mph (1,311 kmph).

Hungarian–British **physicist Dennis Gabor** (1900–79) filed a patent in December for a

January 10 Project Diana radar experiment marks the start of the US space program

February 14 ENIAC computer is introduced at a press conference

July 1 First atomic bomb test at sea

October 19 Genetic recombination in bacteria is described

First proton linear accelerator is produced at Lawrence Berkeley National Laboratory, California

September American chemist Jacob Marinsky (1918–2005) announces isolation of promethium, filling in the last gap in the periodic table

September 21 First anticancer chemotherapy trials described

October 14 Aircraft Bell X-1 flies faster than speed of sound